Sumit Barua

Kalamazoo, Michigan, USA
Email — Homepage — Linkedin — Google Scholar — GitHub

RESEARCH FOCUS

Trustworthy, explainable, and uncertainty-aware AI for high-stakes healthcare, emergency response, and decision-support systems, with emphasis on multimodal and edge healthcare AI, smart-ambulance/EMS reliability, robustness under domain shift, human-AI collaboration, and evidence-grounded LLM/RAG systems.

EDUCATION

Western Michigan University, Kalamazoo, MI Aug 2025 – Present
- M.S. in Computer Science
- Advisor: Prof. Alvis Fong
- Relevant Coursework: Generative AI, Machine Learning, Secure Programming, Cloud Computing, Big Data, MS Thesis

Western Michigan University, Kalamazoo, MI Aug 2018 – Apr 2022
- B.S. in Computer Science
- Relevant Coursework: Java Programming, Data Structures and Algorithms, System Programming, Artificial Intelligence, Software Development

PUBLICATIONS

1. **Barua, S.**, Bahre, R., and Hong, G. Y. "Edge-Enabled Explainable Multimodal AI for Emergency Wound Assessment in Smart Ambulance IoMT Networks." *IEEE Communications Magazine*, vol. 64, no. 9, pp. 86–91, September 2026, doi: 10.1109/MCOM.001.2500767.

2. **Barua, S.**, Bahre, R., Babiceanu, R., Hong, G. "Uncertainty-Aware Wound Localization and First-Aid Decision Support From Smartphone Images Using an Explainable Multimodal AI Framework." *Cureus Journal of Computer Science*, vol. 3, May 18, 2026, es44389-026-00077-z, doi: 10.7759/s44389-026-00077-z.

3. **Barua, S.**, Jabran, S., Rodgers, C., Fong, A. "Quantifying Behavioral Contributions to Diabetes Risk Using an Explainable Multiclass Machine Learning Framework with BRFSS 2024." *Discover Public Health*, vol. 23, Article 1676, 2026. https://doi.org/10.1186/s12982-026-03175-4

UNDER REVIEW & PREPRINTS

1. **Barua, S.**, & Mahmud, B. U. (2026). "Applicability-Aware Reliability Scoring and Prediction of EMS Patient-Care Records for Trustworthy Smart Ambulance AI." *Preprints*, 2026090497. DOI. *Accepted at MDPI AI.*

2. **Barua, S.**, Hong, G., Dursunoglu, H., Rodgers, C., Fong, A. (2026) "CiteGuard-RAG: A Validation-Centered AI System for Evidence-Grounded Question Answering." *arXiv.* arXiv:2609.15830. *Under review at Engineering Reports.*

PEER REVIEW SERVICE

- **Peer reviewer**, *IEEE Communications Magazine*, 2026.
- **Invited Reviewer**, *Communications and Intelligent Networked Computing*, 2026.

SELECTED RESEARCH PROJECTS

Explainable Hybrid Multimodal Framework for Remote Wound Assessment [GitHub]

- Developed an explainable multimodal AI framework using *YOLOv11* for wound localization, *ResNet-50* for anatomical classification, and *Mistral-7B* LLM-assisted generation for first-aid decision support in smart-ambulance and remote-care settings.
- Integrated Grad-CAM-based visual explanation and bounding-box localization to support transparent wound assessment; resulted in two first-author publications in *IEEE Communications Magazine* and *Cureus*.

CiteGuard-RAG: Citation-Constrained System for Legal Question Answering [GitHub]

- Built a retrieval-augmented generation pipeline combining hybrid retrieval, citation-constrained generation, sentence-level validation, and bounded regeneration using *Mistral-7B* and *Sentence Transformers*.
- Evaluated reliability under controlled and external domain-shift settings, achieving 95.4% grounded answer accuracy, 100% citation validity, and a 1.5% hallucination rate.

Explainable Machine Learning for Diabetes Risk Analysis Using BRFSS 2024 [GitHub]

- Trained interpretable machine-learning models on BRFSS 2024 data to classify no diabetes, prediabetes, and diabetes risk using behavioral, demographic, and anthropometric predictors.
- Applied SHAP-based explainability to quantify behavioral contributions to diabetes risk and support transparent public-health decision analysis.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Apr 2026 – Present
WMU Department of Computer Science Kalamazoo, MI

- Develop machine-learning and explainable-AI workflows for research problems involving IoMT systems, smart ambulance networks, healthcare decision support, and cyber-resilient connected systems.
- Develop and maintain Unity-based interactive simulation environments for an NSF-funded research project, supporting experimental workflows, user interaction, and applied AI/game-based system development.
- Deliver guest lectures to different CS classes to familiarize students with our lab's ongoing research projects and experiments.

Software Engineer Jun 2022 – May 2025
Dematic Corporation Grand Rapids, MI

- Developed automation software for order-fulfillment and material-handling systems.
- Supported system integration, testing, troubleshooting, and deployment of reliable industrial automation software.
- Collaborated with cross-functional engineering teams to deliver practical software solutions for warehouse automation environments.

Software Engineering Intern May 2021 – Aug 2021
Frontier Communications Allen, TX

- Designed and implemented a web application using Flask and Jinja to fetch the network port information and its real-time status.
- Developed tools to automate basic troubleshooting based on port status, improving customer service and reducing on-call troubleshooting time for the customer service team.

Undergraduate Learning Assistant Aug 2019 – Dec 2019
WMU Student Suceess Center Kalamazoo, MI

- Supported students in learning Java programming fundamentals, object-oriented programming, algorithmic thinking, and problem-solving techniques.

- Guided students through programming exercises and debugging, helping them identify syntax, runtime, and logical errors.

TECHNICAL SKILLS

Programming:	Python, SQL, Java, C
Machine Learning / AI:	scikit-learn, TensorFlow, XGBoost, YOLOv11, ResNet-50, SHAP, Grad-CAM, Retrieval-Augmented Generation, Large Language Models
Development Tools:	Git, Linux, Django, Flask, Streamlit, REST APIs, MS SQL Server, Unity
Research Methods:	Dataset preprocessing, feature engineering, model evaluation, explainability analysis, domain-shift evaluation, uncertainty-aware decision support

HONORS AND AWARDS

- **Global Education Graduate Merit Scholarship Recipient**
  Western Michigan University *Fall 2025–Spring 2027*
  - Awarded $10,000 in renewable tuition support over four semesters based on academic merit.

- **Haenicke Global Education Undergraduate Merit Scholarship Recipient**
  Western Michigan University *Fall 2018–Spring 2022*
  - Awarded $16,000 in renewable merit-based tuition support throughout eight semesters of undergraduate study.

LEADERSHIP AND COMMUNITY SERVICE

- International Orientation Leader, Western Michigan University 2020
- Fall Welcome Student Lead, Western Michigan University 2020
- Volunteer, Kalamazoo Loaves & Fishes 2021